Binary Change Detection on EO-SAR Image Pairs

Abstract

In this study, a deep learning model has been developed for binary classification of change detection from pre-event and post-event EO-SAR satellite images. This model will classify each pixel into either a changed or an unchanged area. The proposed solution utilizes a Siamese ResNet34 U-Net framework, which learns the difference between the features of two paired images. In this work, the original labels of four categories have been converted to binary labels according to the requirements of the assignment. The total size of the provided training data for the assignment is about 13.6 GB. Since the computational and memory limitations of Kaggle restrict the usage of complete data, only a subset of training data (~3.91 GB) has been selected for training the model. Likewise, ~1.63 GB and 385 MB of the provided validation and testing data, respectively, have been used for developing and evaluating the model.

Literature Survey

Change detection in remote sensing is used in disaster assessment, urban monitoring and environmental surveillance. Old methods used thresholding and pixel differencing. These methods failed under noise and real world conditions. Ronneberger et al. (2015) built U-Net for medical image segmentation. It has an encoder decoder structure with skip connections. Skip connections help the decoder recover fine spatial details. It later became a standard architecture for remote sensing segmentation tasks [1].Daudt et al. (2018) adapted U-Net into a Siamese network for change detection. Two encoder branches with shared weights process pre event and post event images separately. Their features are compared to produce a binary change map. This became a strong baseline for bi-temporal change detection [2]. Bandara and Patel (2022) proposed ChangeFormer. It places a vision transformer inside a Siamese framework. It captures long range spatial dependencies that CNNs cannot. It performs well but needs more GPU memory and larger training data [3].Cao et al. (2024) proposed Siam-Swin-UNet. It combines a Swin Transformer with an encoder decoder design. It improved boundary detection and reduced false positive predictions in high resolution imagery [4].Based on this survey a Siamese ResNet34 U-Net was selected for this work. It gives a good balance between accuracy and training feasibility under limited GPU resources. A combined BCE and Dice loss was used

Methodology

1. Problem Formulation

The objective of this work is binary change detection using paired pre-event and post-event EO-SAR satellite imagery. The model predicts a pixel-level binary segmentation mask representing changed and unchanged regions. As specified in the assignment, the original four semantic annotation classes were remapped into two classes:

- Background and Intact → No-Change (0)

- Damaged and Destroyed → Change (1)

This remapping was applied consistently across training, validation, and test splits before training and evaluation.

2. Data Preprocessing

Before training, several preprocessing operations were applied:

- Images and masks were resized to 256×256
- Input images were normalized using ImageNet normalization statistics
- Binary masks were generated using the mandatory label remapping strategy
- Masks were converted into single-channel binary segmentation maps

3. Data Augmentation

To improve model robustness and reduce overfitting, online data augmentation techniques were applied during training:

- Horizontal flipping
- Vertical flipping
- Random 90° rotation

4. Model Architecture

A Siamese ResNet34 U-Net architecture was used for binary change detection. The network processes pre-event and post-event images using two parallel encoder branches and learns feature differences between the two temporal states.

4.1 Siamese Encoder

The encoder backbone was based on ResNet34 pretrained on ImageNet. Two encoder branches extracted hierarchical feature representations from the pre-event and post-event images separately. Absolute feature differences between corresponding encoder layers were computed to capture temporal changes.

Residual learning in ResNet34 improved feature extraction capability while maintaining stable optimization for deeper networks.

4.2 U-Net Decoder

The decoder followed a U-Net-style architecture with progressive upsampling and skip connections. The decoder reconstructed high-resolution segmentation masks from multi-scale feature difference maps generated by the Siamese encoder.

The final output layer produced a single-channel binary change probability map.

5. Loss Function

A hybrid loss function combining Weighted Binary Cross-Entropy (BCE) Loss and Dice Loss was used.

The combined loss function is defined as:

$$L = 0.5 \times L_{BCE} + 0.5 \times L_{Dice}$$

6. Optimization Strategy

The model was trained using the AdamW optimizer with weight decay regularization.

Training configuration:

- Optimizer: AdamW
- Initial learning rate: 1×10^{-4}
- Batch size: 4
- Image size: 256×256
- Number of epochs: 50
- Learning rate scheduler: ReduceLROnPlateau
- Early stopping based on validation F1-score

7. Evaluation Metrics

The model was evaluated using the metrics specified in the assignment:

- Intersection over Union (IoU)
- Precision
- Recall
- F1-score
- Confusion Matrix

8. Computational Constraints

The complete training dataset size was approximately 13.6 GB. Due to storage and GPU memory limitations in the Kaggle environment, a reduced subset of the training data (~3.91 GB) was used for experimentation. Approximately 1.63 GB of validation data and 385 MB of test data were also utilized.

Results and Discussion

The Siamese ResNet34 U-Net model was trained for binary EO-SAR change detection using pre-event and post-event satellite image pairs. The model was trained using weighted Binary Cross-Entropy and Dice loss to handle class imbalance in the dataset. The best model checkpoint was obtained at epoch 27 with a threshold value of 0.90.

Best Validation Results

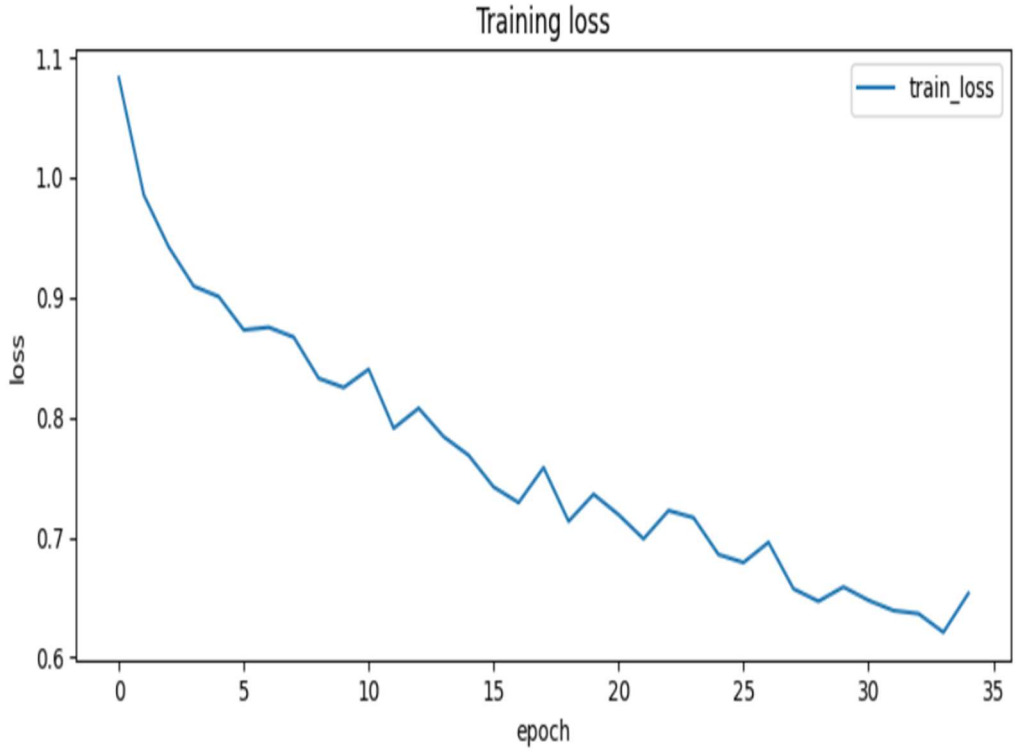
Metric	Score
Accuracy	0.9810
Precision	0.5925
Recall	0.4340
F1-score	0.5010
IoU	0.3342

Final Test Results

Metric	Score
Accuracy	0.9764
Precision	0.0347
Recall	0.0792
F1-score	0.0483
IoU	0.0247

The confusion matrix on the test dataset was:

- TP: 3014
- TN: 4,924,378
- FP: 83,833
- FN: 35,047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epoch
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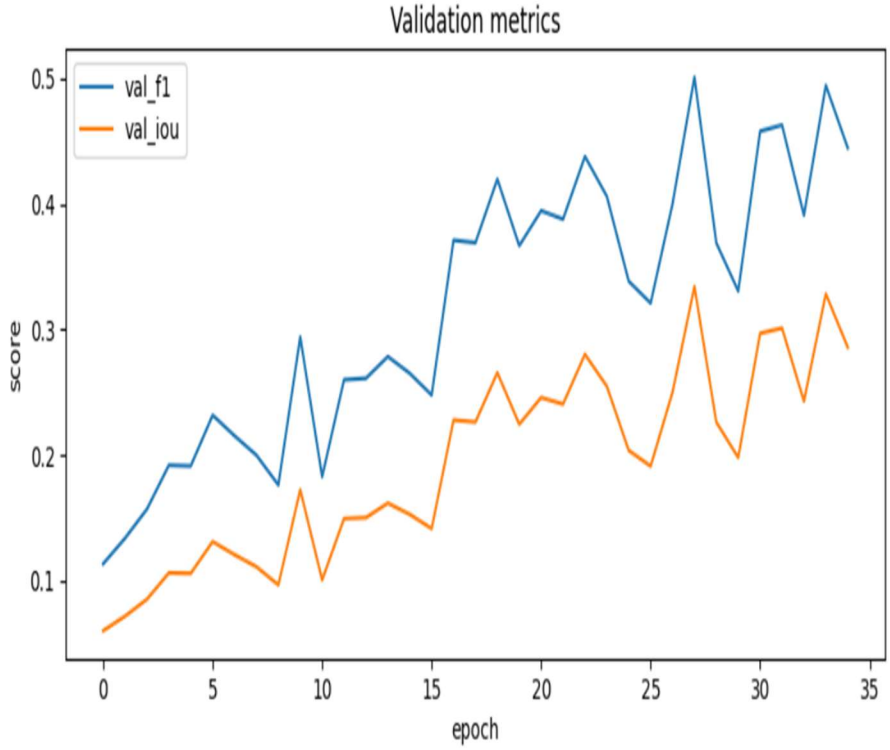
Validation metrics
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Draft Session
GPU T4 ×2 On

Session	Disk
1h:23m	406.9MiB
12 hours	Max 57.6GiB

CPU

CPU	RAM
0.00%	8.2GiB
	Max 30GiB

GPU

GPU	GPU Memory
0.00%	1.4GiB
	Max 15GiB

GPU

GPU	GPU Memory
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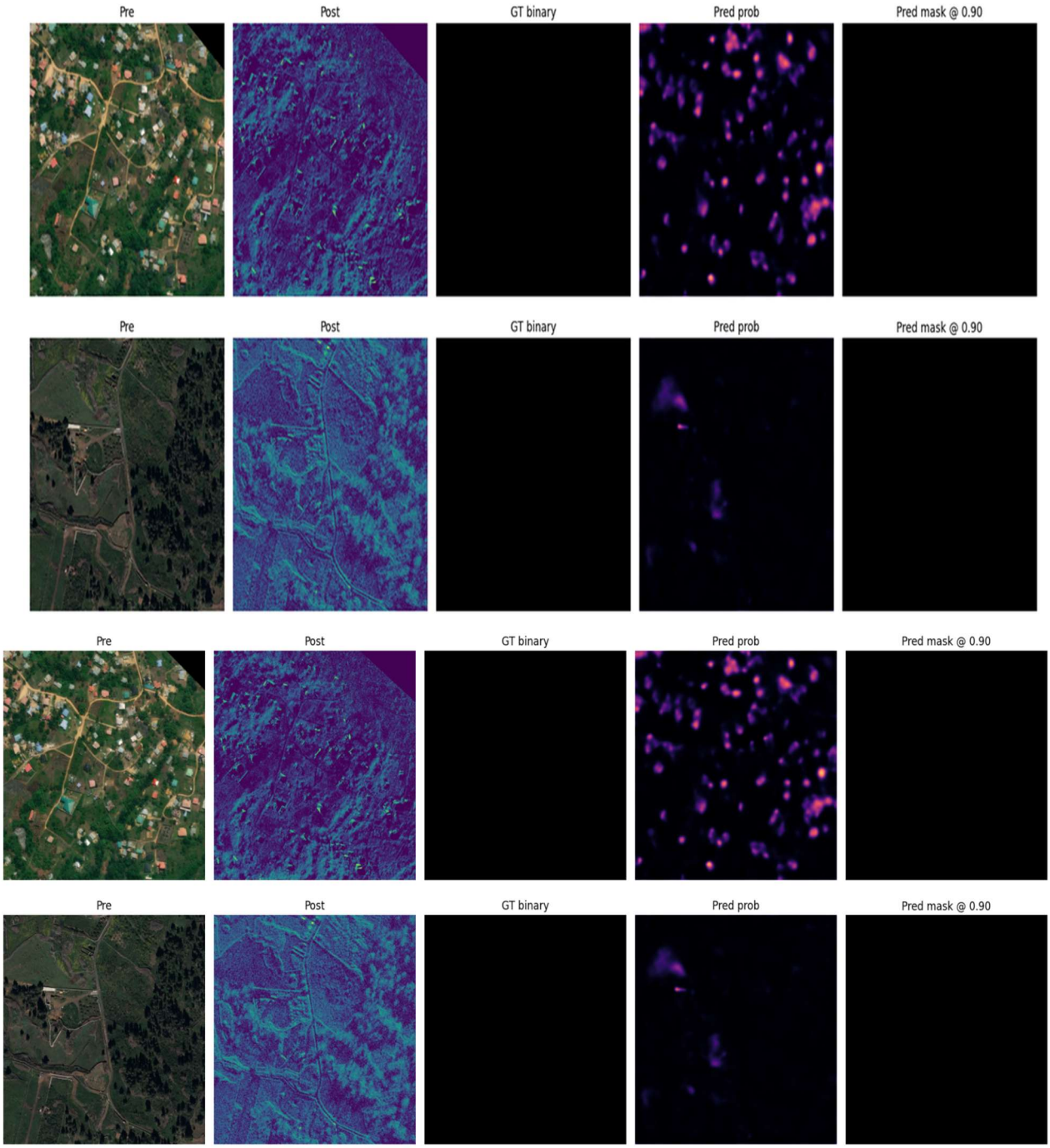
	Max 15GiB


Pre
Post
GT binary
Pred prob
Pred mask @ 0.90
Pre
Post
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Pred prob
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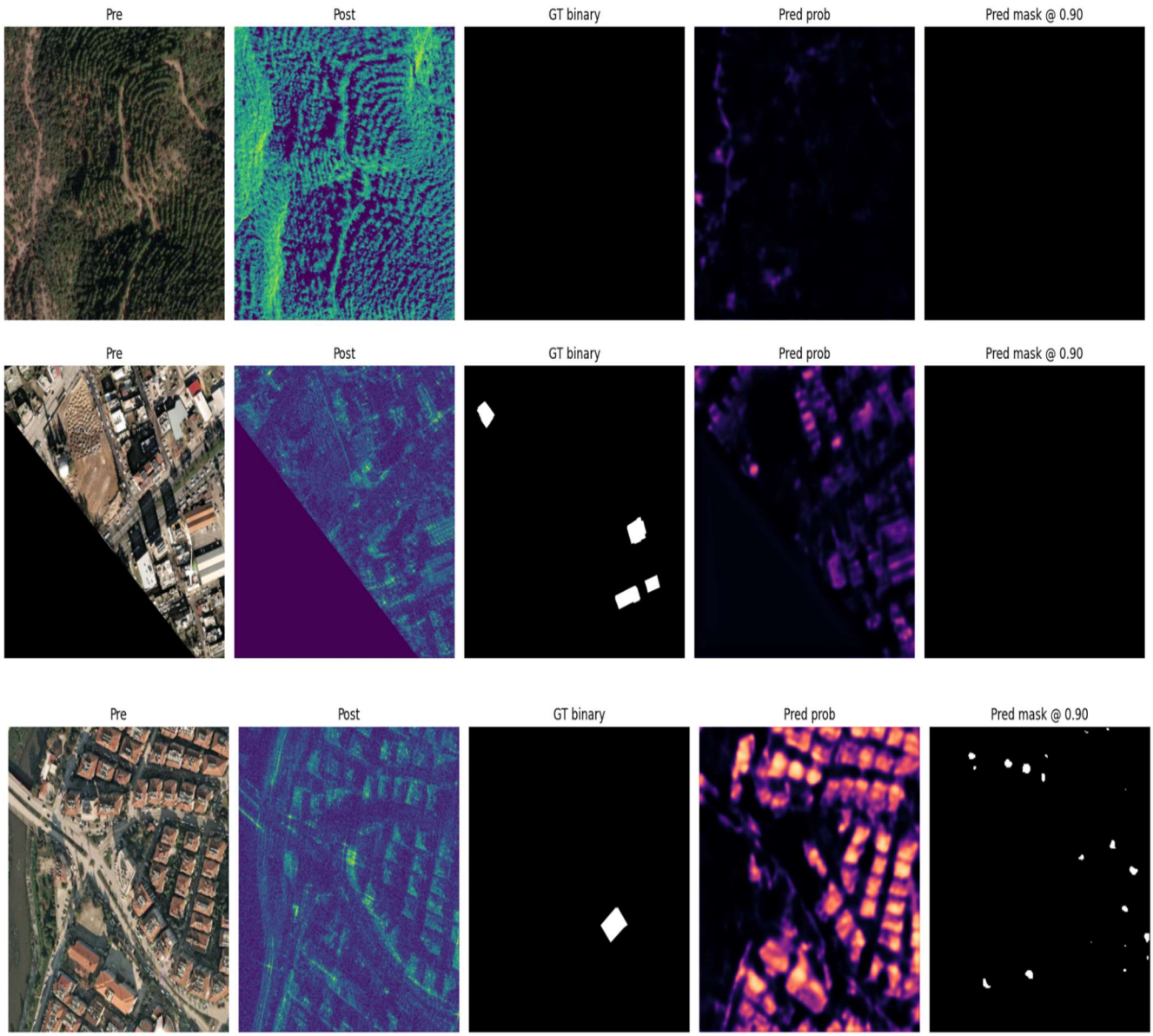

Future Work

1. Train on complete dataset Only a small subset was used due to Kaggle GPU limits. Training on the full 13.6 GB dataset will improve model generalisation across different disaster types and scenes.
2. Try transformer based architecture ChangeFormer and Swin-UNet capture long range dependencies better than CNNs. These architectures would help detect changes in complex scenes where CNN features are not enough.
3. Stronger data augmentation Only basic flipping and rotation were used. Adding colour jitter, random cropping, and SAR specific noise augmentation would make the model more robust to sensor variations.

4. Higher image resolution Images were resized to 256x256 due to memory limits. Training at 512x512 or higher would preserve more spatial detail and improve boundary detection.

Conclusion

In this work, a Siamese ResNet34 U-Net model was used for binary EO-SAR change detection using pre-event and post-event satellite images. The model learned temporal differences between image pairs and generated binary change masks. Weighted Binary Cross-Entropy and Dice loss were used. The model achieved reasonable validation performance but lower test performance due to dataset imbalance, EO-SAR modality differences, and limited training data caused by computational constraints.

References

1. Ronneberger, O., Fischer, P., & Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. *MICCAI 2015.* arXiv:1505.04597
2. Daudt, R. C., Le Saux, B., & Boulch, A. (2018). Fully Convolutional Siamese Networks for Change Detection. *ICIP 2018.*
3. Bandara, W. G. C., & Patel, V. M. (2022). A Transformer-Based Siamese Network for Change Detection. *IGARSS 2022.* arXiv:2201.01293
4. Cao, et al. (2024). A Siamese Swin-UNet for Image Change Detection. *Scientific Reports.* DOI:10.1038/s41598-024-54096-8